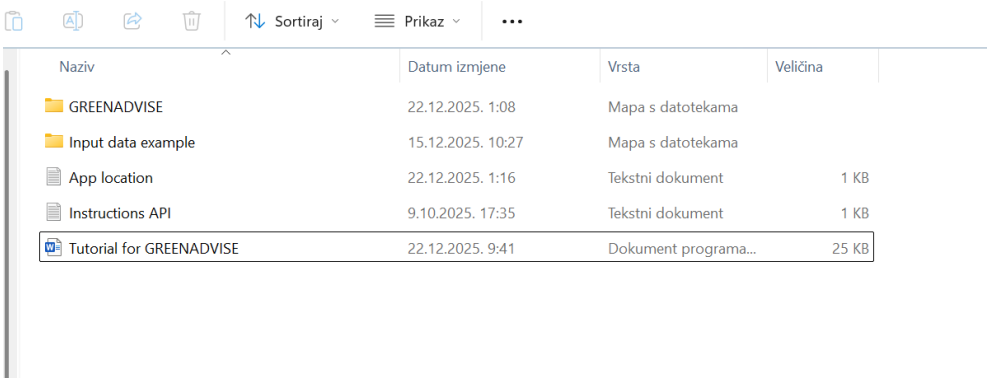
GREENADVISE

--- tutorial ---

1. *First steps*

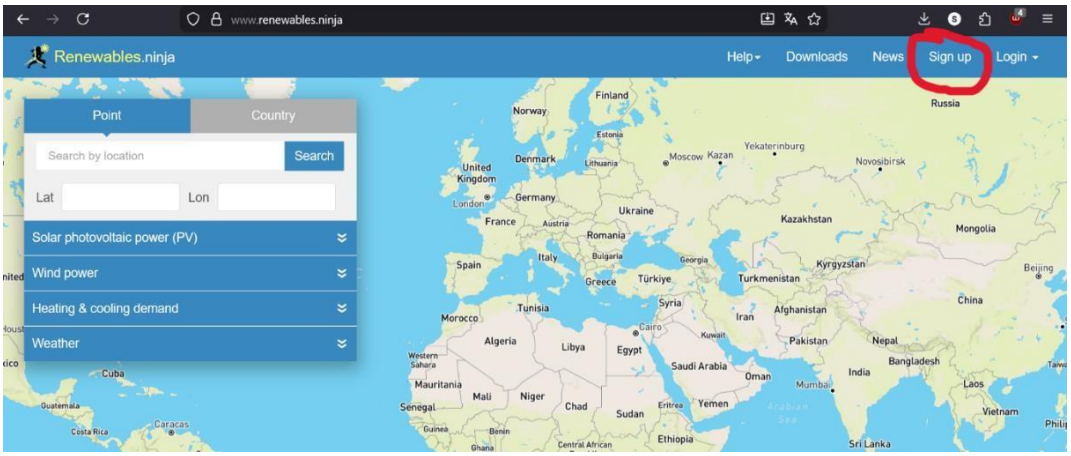
After downloading **GREENADVISE_v4.zip**, unpack it in the folder of your destitnation.

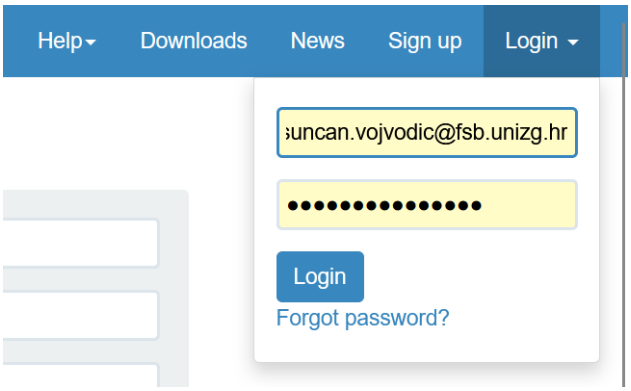


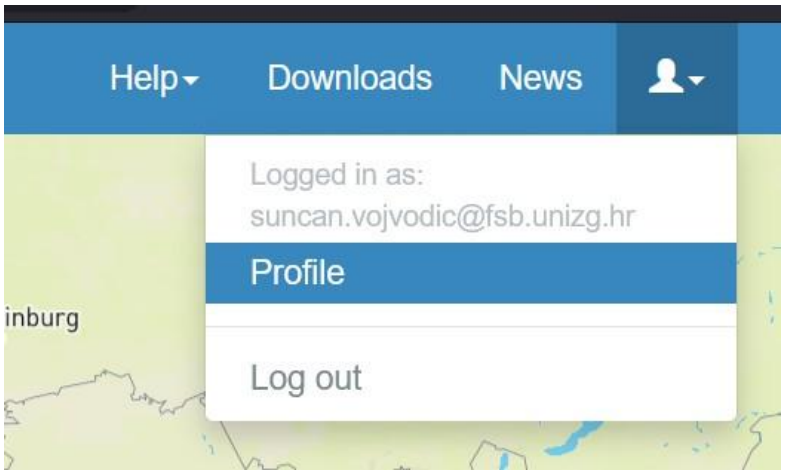
Under **App location** you can find where .exe file is located: **GREENADVISE\dist\GREENADVISE.exe**.

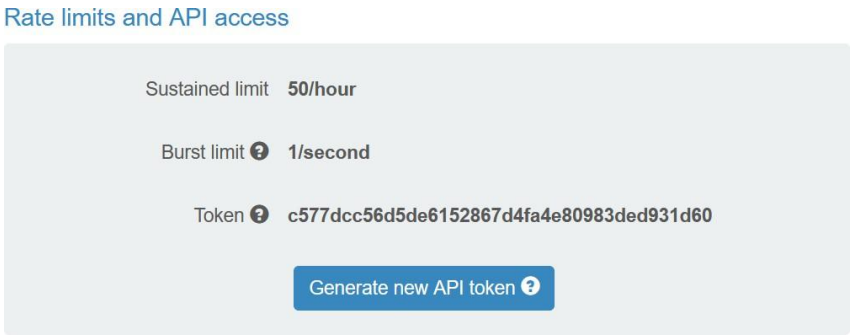
Under **Instructions API** are instructions of creating your account on https://www.renewables.ninja/ for API key, which is needed for fetching weather database.

- Go to: https://www.renewables.ninja/
- Click **Sign up** and **Register**
- After registration **Login**
- Go to the top right and click profile
- Your API key is visible under Rate limits and API access as **Token**





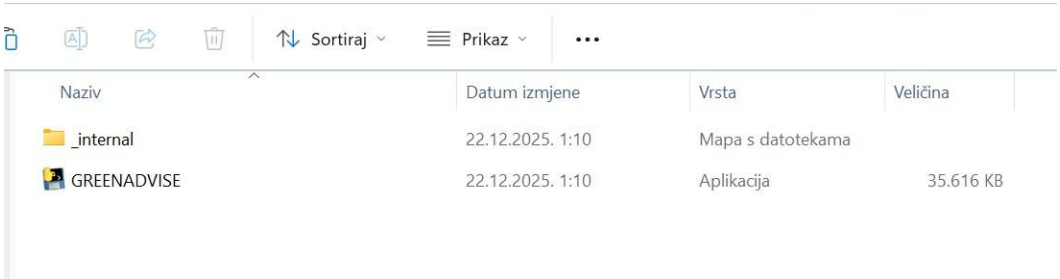




Your API limit will have 50 calls per hour. If you spent and can not wait for the refresh, you can Generate new API token.

2. Second steps

Run GREENADVISE.



If for some reason the app won't open, you can call it manually in terminal under> **cd "Your_Location\GREENADVISE\dist\GREENADVISE"** and typing **GREENADVISE.exe**.

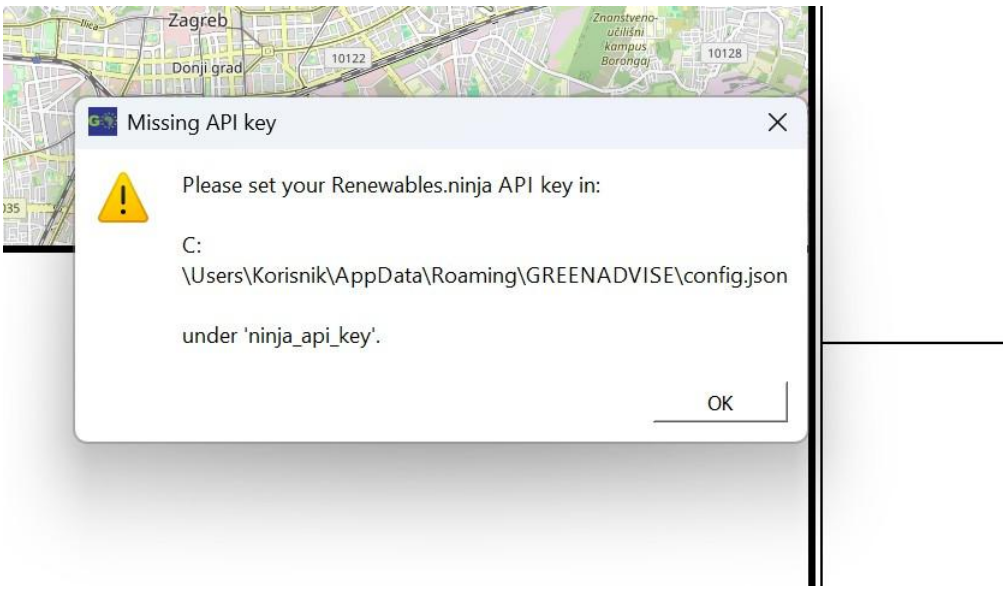
```
C:\Users\Korisnik>cd "C:\Users\Korisnik\Desktop\My folder\GREENADVISE\dist\GREENADVISE"

C:\Users\Korisnik\Desktop\My folder\GREENADVISE\dist\GREENADVISE>GREENADVISE.exe
```

Or you can open trough the main script called **main_app.py** through terminal or any supporting **Python IDE** (e.g., Visual Studio Code, Spyder, etc.).

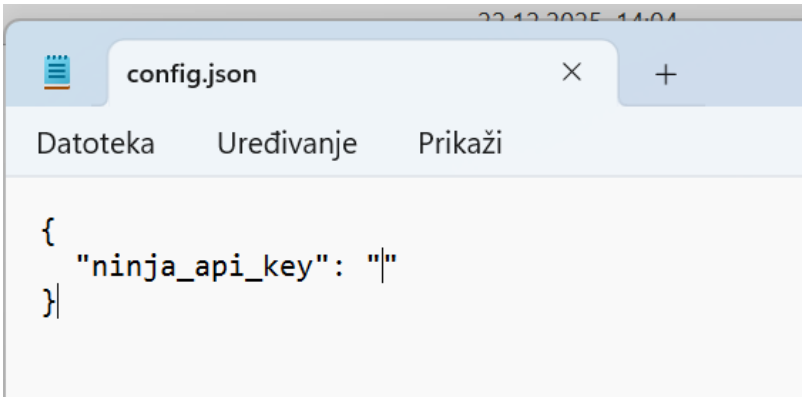
3. Third steps

Follow **Instructions API** notes.

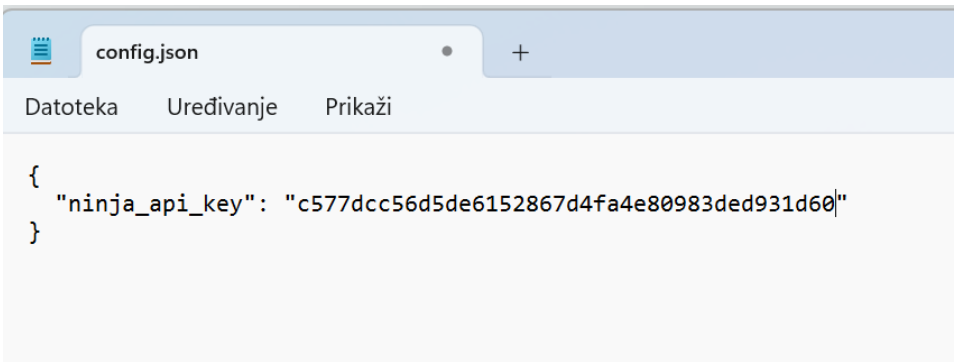
When you first open GREENADVISE, when clicked on fetching data (e.g., Generation technology) you will get popup button saying **where** to input API key:



You must search the path in your folder for **config.json** file and open it.



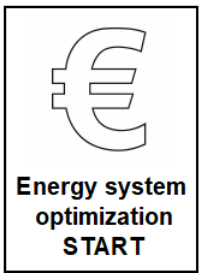
Between quotation marks **input your Renewables ninja API key**.



Save it. **Restart the GREENADVISE**.

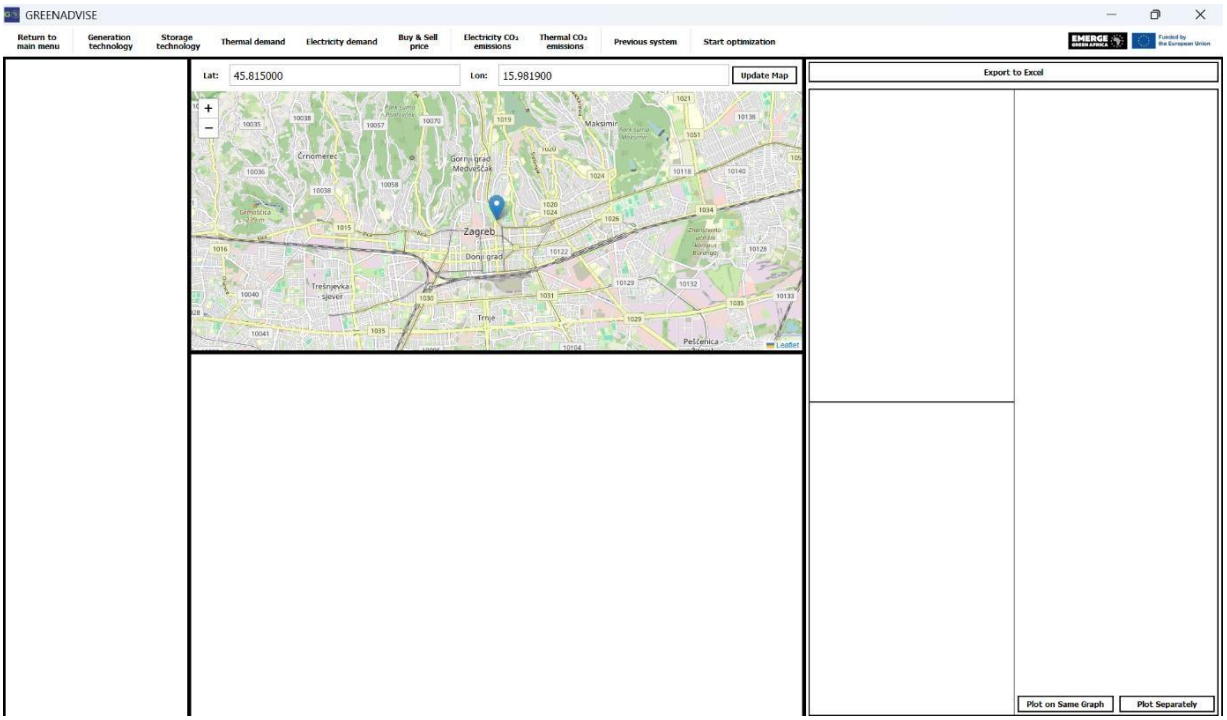
4. Fourth steps

GREENADVISE



Enter the START.

(sometimes app doesn't respond in time. You can restart it or wait a couple of seconds before clicking START).

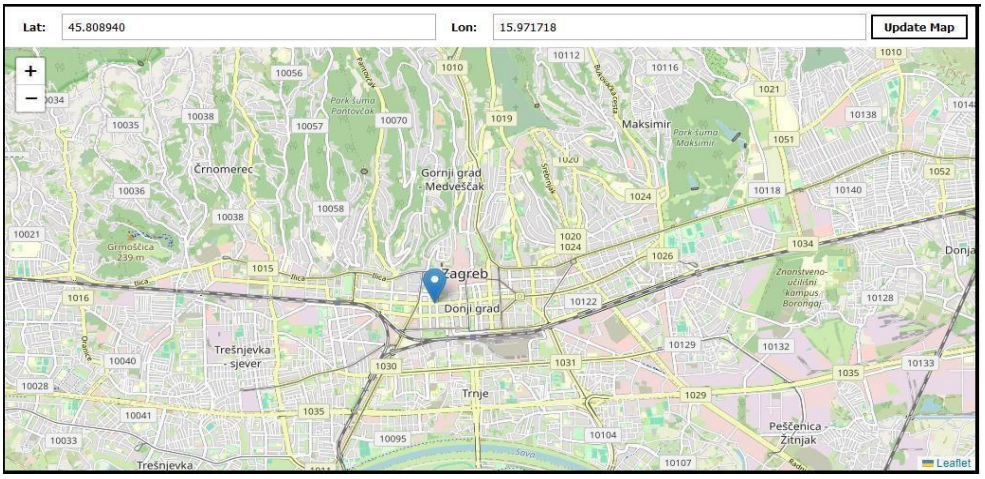


Main interface of the tool. Hovering over black lines, you **can narrow down** or **expand sections**.

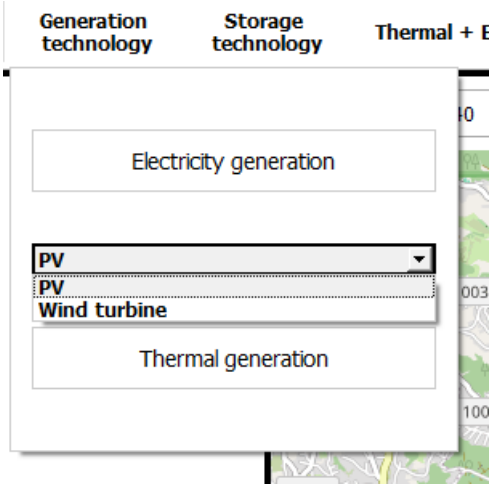
In GREENADVISE you have options for Generation technology (both electricity and thermal), Storage technology, Thermal and electricity demand, Buy & sell price, CO_2 for electricity and thermal emissions.

5. Fifth steps – testing the tool

Pick geographical location with **pulling** the location marker or inputting latitude and longitude data by clicking **Update Map**.



On top bar you have options for choosing your technologies in '**Generation technology**' and '**Storage technology**'.



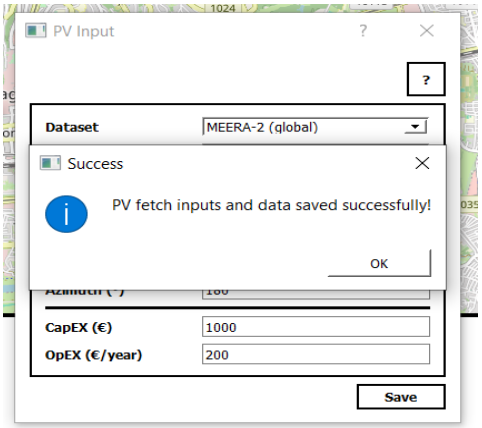
You can choose for your PV and Wind turbine **3** different ways in:

1) Simulating (uses theory formulas for output power)

2) Upload (see .zip folder Upload_data_examples in GREENADVISE folder)

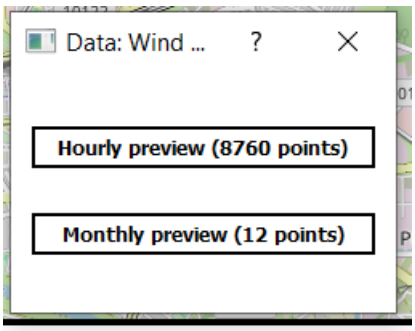
3) Fetch (uses Renewables.ninja logic)

You input your technologies by '**Save**' and you will get confirmed message.

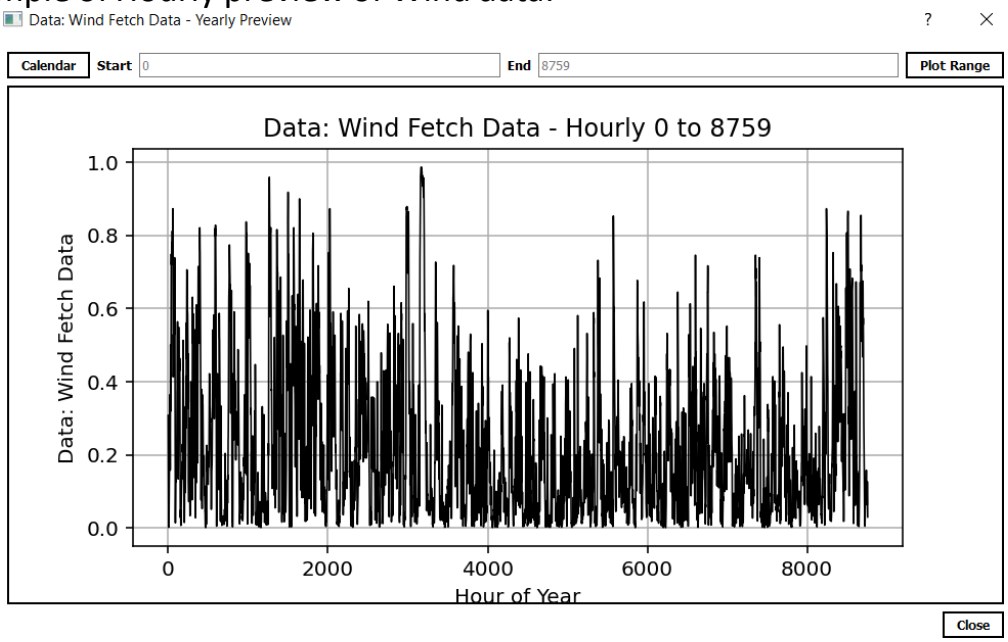


***Azimuth data is in the range from 0-360. In renewables.ninja, an azimuth of 180° always corresponds to an equator-facing orientation: south-facing in the Northern Hemisphere and north-facing in the Southern Hemisphere.**

After you inputed your technologies, you can **preview** (or delete) them on the left side menager by clicking Input or plot them by clicking Data:

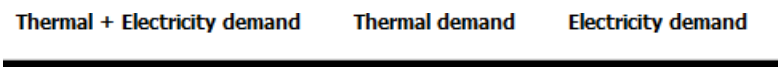


Here is an example of Hourly preview of Wind data:



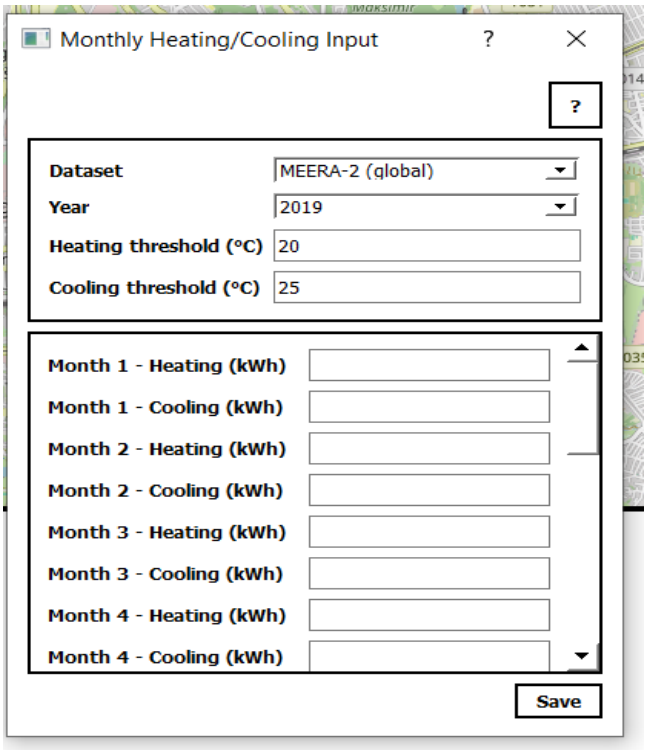
You can plot range from hourly values by clicking Plot range, or via Calendar by choosing Start and End Date.

Then you will need to input your **Demands**:



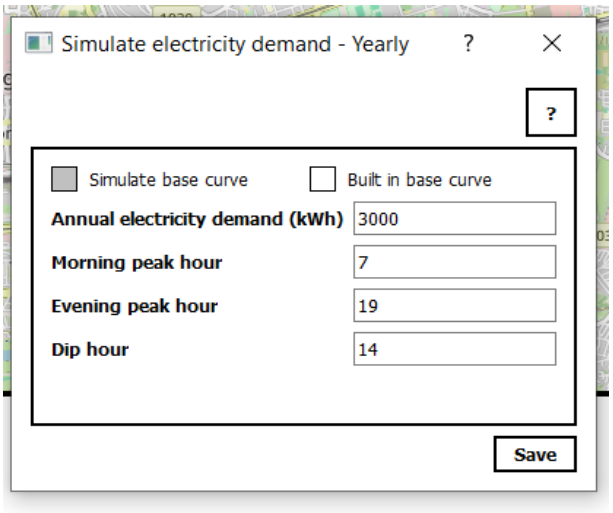
1) **Thermal + Electricity demand** lets you input both electricity demand and thermal demand in **Base power**, and adjusting it with parameters.

2) **Thermal demand** gives you **2** option for inputting, via Upload (see Upload_data_examples) or Simulating heating and cooling demand in Yearly or Monthly option.

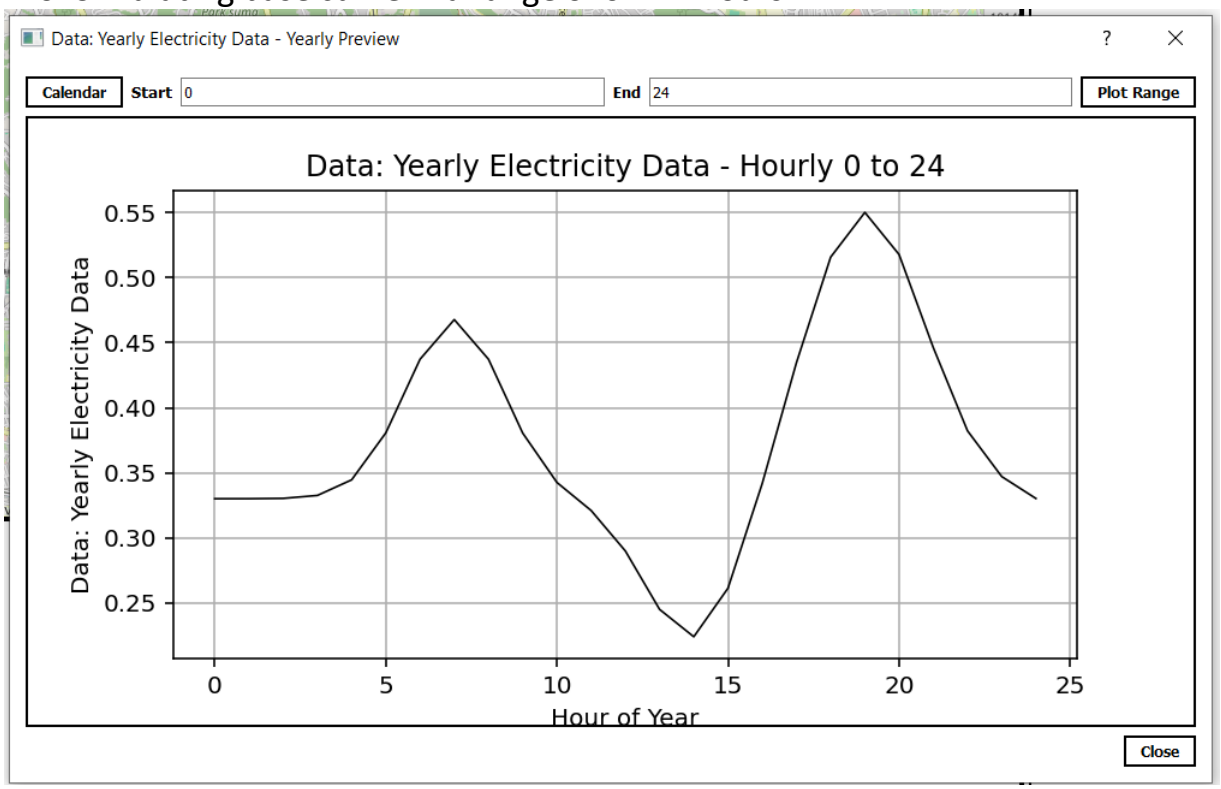


Option for **Monthly Heating/Cooling Input.**

3) **Electricity demand** gives you the same option as **Thermal demand** with exception of choosing daily power load curve with option of **Simulate** and choosing already **Built in base curve**.



Preview of Simulating base curve in a range of 0 – 24 hours:



Input your electricity price in **Buy & Sell price**. You can input your price with options:

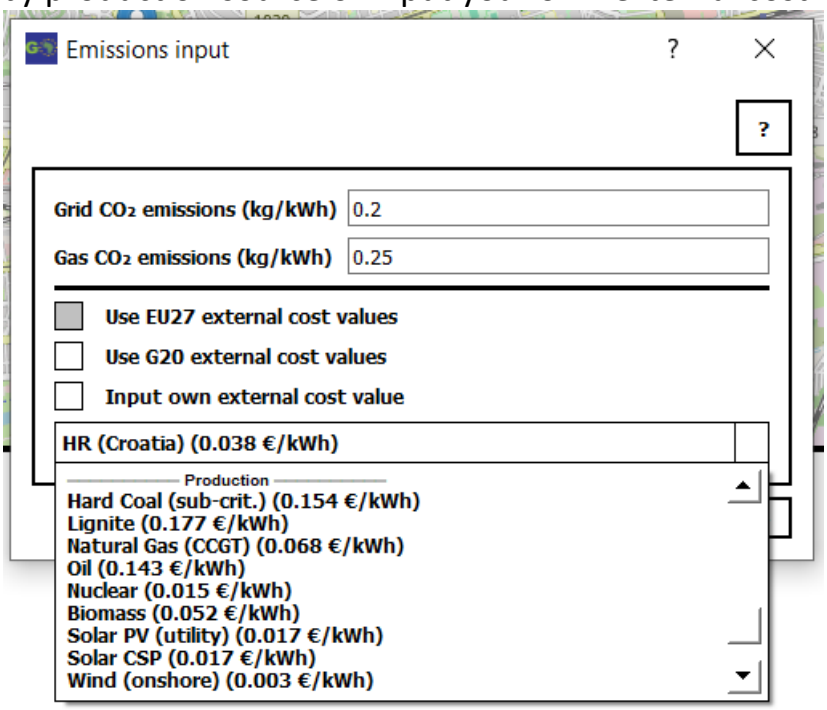
4) Price select by country (**5** countries electricity data from ENTSO-E day ahead price)

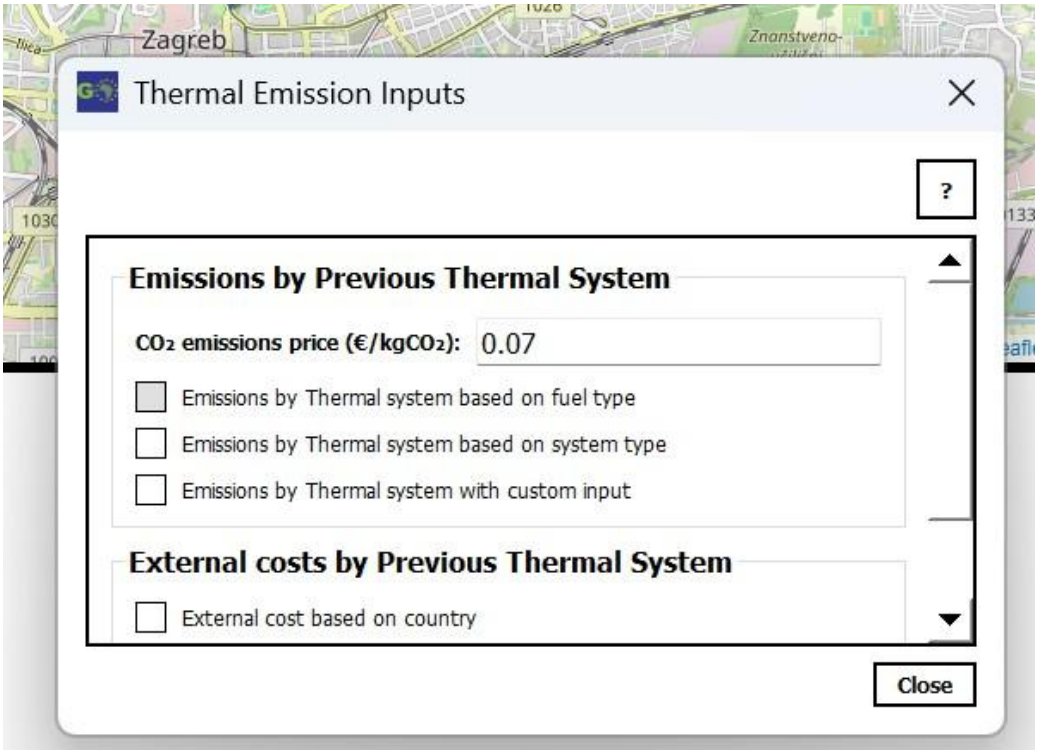
5) **Dual** tariff system

6) **Single** tariff system

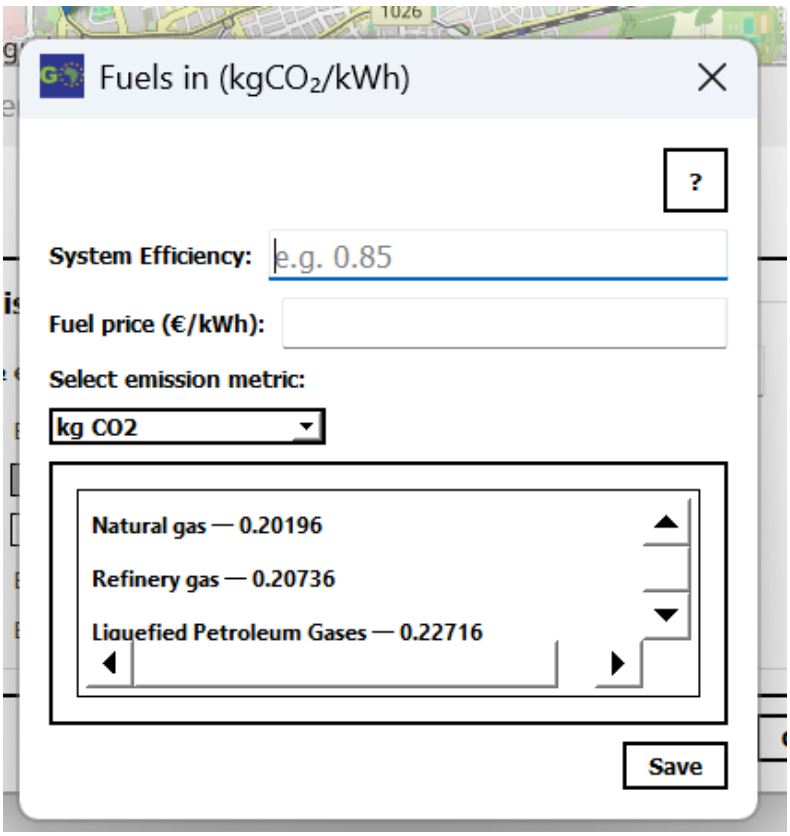
7) **Upload** price data (again, see examples on Upload_data_examples)

You can also input your **CO_2 emission** data for stake in Grid emission (kg/kWh) and Gas emission (kg/kWh). You can also choose **external cost** from emissions based on choosed countries, by production source or input your own external cost value.

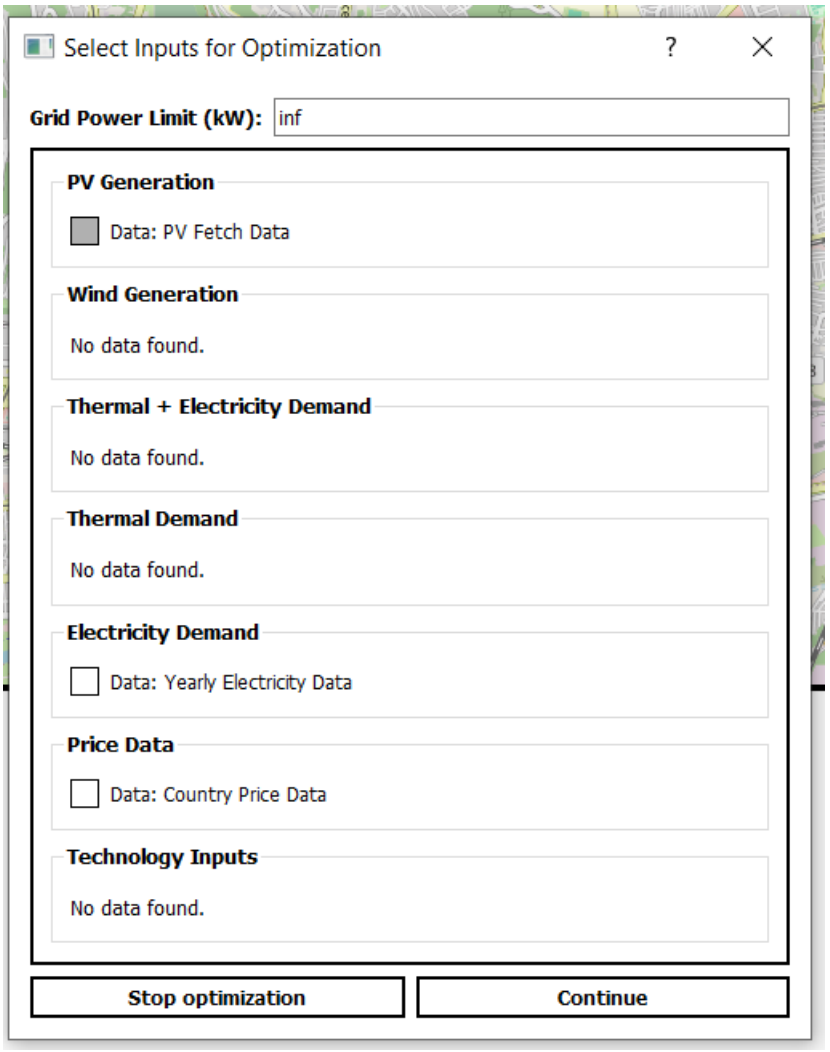




In Thermal Emission Inputs you can input your previous (base) heating fuel and price, as well as CO_2 emission stake and price; and external cost for CO_2 thermal.



After your choosed inputs (minimal inputs needs to be e.g., at least one generation technology (e.g., PV), Electricity demand, Buy & Sell price) you can click **Start optimization**.



Grid Power Limit (kW) lets you limit withdrawal of electricity power from the grid (if **inf**, you can withdrawal infinite).

You can choose between deterministic (one year) and stochastic (five years) optimization. If you choose upload option, you can also choose different optimization based on your upload data.

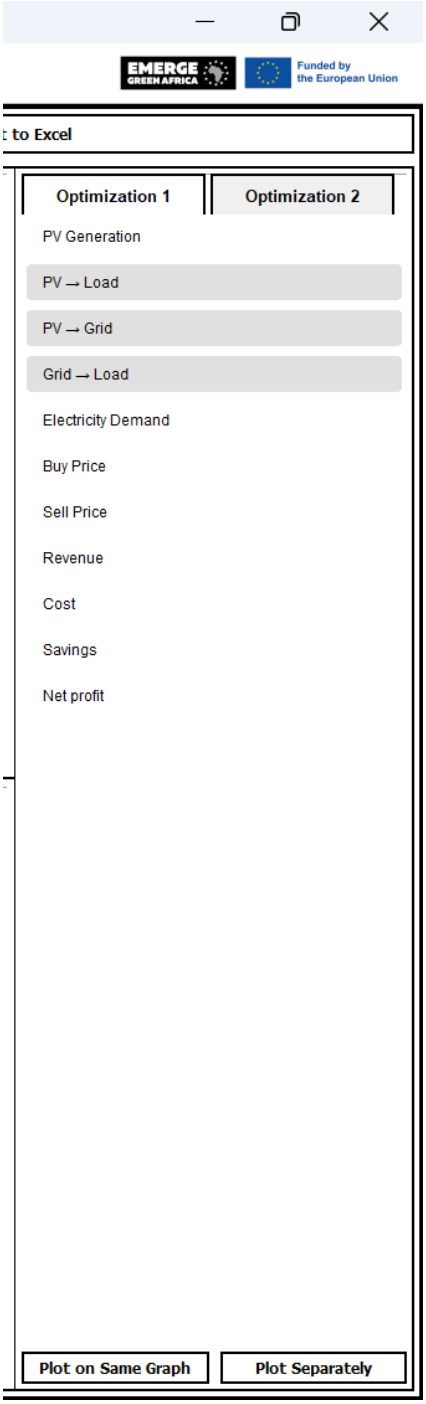
You need to check all the boxes you want for your optimization to continue, see first box. After checking all the boxes, click **Continue**.

Optimization progress last up to 10 seconds.

After optimization ends you can see on the first column on the right prints of Optimization, your **financial** part. Below financial column, you get prints of results of **CO_2 emissions** and **external cost** savings. On the right column you can plot **energy flows** on the same graph or separately (in hourly and monthly previews as explained before.

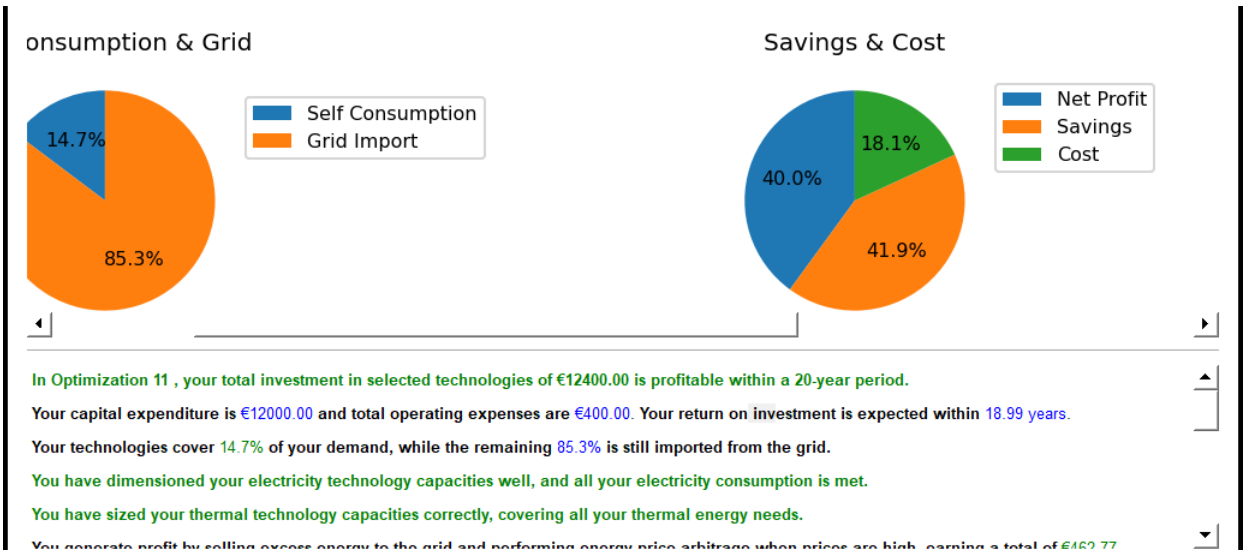
<u>Optimization gives results on the span of 20-years based on optimized one year. Battery CAPEX are counted twice because of the life expectancy.</u>

On the right menager you can plot on the same graph or separately, and can preview monthly or hourly, and can Plot range or choose timeframe from Calendar.

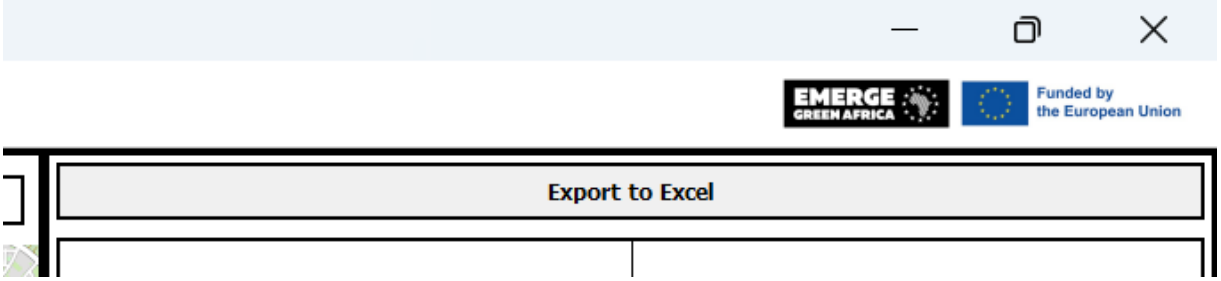


Plotting some energy flows on same graph of first 24 hours.

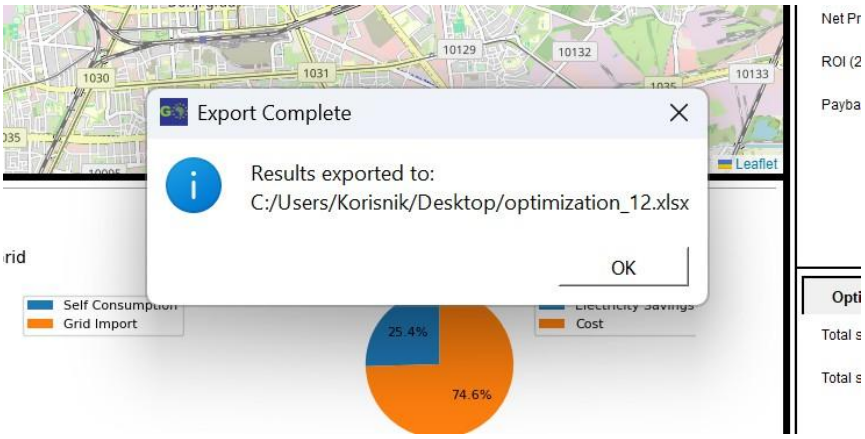
In optimization tab below map, you get **textual** results and **pie charts** (self-demand balance, financial balance, CO_2 balance) based on optimization results. If your results don't satisfy payback return or demand, text will be red and instruction how to overcome will be printed. If results are positive, your text will be green. Textual results and pie charts are there to educate and present results based on inputs in intuitive way.



Finally, you can export your optimization by clicking **Export to Excel**, and see energy flows in numbers and sectors.



When clicking Export to Excel, you need to choose your path and wait for a couple of seconds to get conformation message:



In excel file you can deep dive into results and hourly energy flow.

Međuspremnik | Font | Poravnanje | Broj | Stil

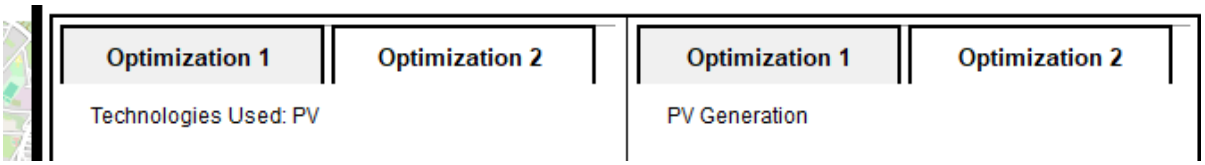
A1 | Hour

	A	B	C	D	E	F	G
1	Hour	PV Generation	PV → Load	PV → Grid	PV → Battery	PV → Heat Pump (heating)	PV → Heat Pump (cooling)
2	0	0	0	0	0	0	0
3	1	0	0	0	0	0	0
4	2	0	0	0	0	0	0
5	3	0	0	0	0	0	0
6	4	0	0	0	0	0	0
7	5	0	0	0	0	0	0
8	6	0	0	0	0	0	0
9	7	0,604	0	0,604	0	0	0
10	8	2,909	0,4595	2,2683	0	0,1812	0
11	9	4,444	0,3551	3,9183	0	0,1706	0
12	10	5,449	0,2753	5,0098	0	0,1638	0
13	11	4,616	0,2836	4,1701	0	0,1624	0
14	12	3,456	0,2795	3,0089	0	0,1677	0
15	13	2,526	0,2753	2,0754	0	0,1753	0
16	14	1,431	0,2671	0,9698	0	0,194	0
17	15	0	0	0	0	0	0
18	16	0	0	0	0	0	0
19	17	0	0	0	0	0	0
20	18	0	0	0	0	0	0
21	19	0	0	0	0	0	0
22	20	0	0	0	0	0	0
23	21	0	0	0	0	0	0
24	22	0	0	0	0	0	0
25	23	0	0	0	0	0	0

Optimization | Information | Financial Summary | Emissions Summary | PV Generation | Thermal ...

Scroll through the sheets for different theme of the optimization.

For each pop-up window there are **question mark** buttons. You can also start **multiple optimizations** in the same program.



In the right menager you need to **select what** number of optimization you want to download for excel. (e.g., you want to download optimization_2, you need to click Optimization 2):

NEXT STEPS:

- Test the GREENADVISE tool
- If you have trouble, **feel free to send questions on e-mail**: suncan.vojvodic@fsb.unizg.hr with subject GREENADVISE.

In conclusion, GREENADVISE tool offers hourly optimal energy flow calculations with two optimization methods, deterministic and stochastic, enableing Users to calculate optimal technologies based on their Case; based on Users input of location, selected technology, and Users energy demand. GREENADVISE offers results in educational, technical, environmental and financial indicators, presenting them in textual, numerical and visual form.